TF (Term Frequency) and IDF (Inverse Document Frequency)

1. Term Frequency (TF)

TF measures how frequently a term appears in a document. It is calculated as:

$$TF(t,d) = \frac{\text{Number of times term } t \text{ appears in document } d}{\text{Total number of terms in document } d}$$

2. Inverse Document Frequency (IDF)

IDF measures how important a term is by considering how many documents contain the term. It is calculated as:

$$IDF(t) = \log\left(\frac{N}{df(t)}\right)$$

Where:

- N = Total number of documents in the corpus
- df(t) = Number of documents containing the term t

A higher IDF means the term is rare across the corpus, making it more important.

Example: Calculating TF and IDF

Corpus (Collection of Documents)

Let's consider a small corpus with three documents:

Doc 1: "The cat sat on the mat"

Doc 2: "The dog barked at the cat"

Doc 3: "The cat chased the mouse"

Step 1: Compute TF for the term "cat" in each document

The total number of words in each document:

- **Doc 1:** 6 words
- **Doc 2:** 6 words
- **Doc 3:** 5 words

Now, count occurrences of "cat":

- **Doc 1:** appears **1** time → $TF = \frac{1}{6} = 0.1667$
- **Doc 2:** appears **1** time → $TF = \frac{1}{6} = 0.1667$
- **Doc 3:** appears **1** time → $TF = \frac{1}{5} = 0.2$

Step 2: Compute IDF for "cat"

- The word "cat" appears in **all 3 documents** → df(cat)=3
- Total number of documents N=3

Using the IDF formula:

$$IDF(\text{cat}) = \log\left(\frac{3}{3}\right) = \log(1) = 0$$

Since IDF is **0**, it means "cat" appears in all documents, making it less significant for distinguishing between them.

Conclusion

TF helps measure term importance within a document, while IDF helps determine how unique the term is across documents. If a word appears in every document, its IDF is **0**, meaning it's not useful for distinguishing between documents.

How TF and IDF Help in Text Representation

TF (Term Frequency) and IDF (Inverse Document Frequency) are essential for **vectorizing text data**, making it useful for machine learning models, search engines, and NLP applications. Together, they form the **TF-IDF score**, which helps in the following ways:

1. **Weighting Important Words**
   - Words that appear frequently in a document but rarely in the entire corpus get a **higher weight**.
   - Common words (like "the," "is") that appear in all documents get a **lower weight**.
2. **Improving Search Engine Ranking**

    - Search engines use TF-IDF to rank pages.
    - A query term with a high **TF-IDF score** in a document means the document is **more relevant** to the query.
3. **Feature Extraction for Machine Learning**
    - TF-IDF converts raw text into numerical form for classification, clustering, and NLP models.
4. **Summarization & Keyword Extraction**
    - Words with high **TF-IDF scores** are often **key topics** of a document.

What Can We Infer from TF and IDF?

1. **High TF & High IDF → Important & Unique Word**
    - Example: "COVID-19" in a medical article.
    - Meaning: This word is **crucial for document meaning**.
2. **High TF & Low IDF → Common Word**
    - Example: "the," "is," "and".
    - Meaning: Appears frequently in all documents, **not useful** for distinguishing documents.
3. **Low TF & High IDF → Rare but Significant Word**
    - Example: "Neural Network" in an AI research paper.
    - Meaning: May be **highly relevant** for categorizing the document.
4. **Low TF & Low IDF → Unimportant Word**
    - Example: "however" or "thus" in academic papers.
    - Meaning: Not significant in distinguishing between documents.

Example of TF-IDF in Action

Corpus (Collection of Documents):

1. **Doc 1:** "The sun is shining bright"
2. **Doc 2:** "The weather is bright and warm"
3. **Doc 3:** "The sun and the weather are beautiful today"
4. **Doc 4:** "Bright sunlight makes the day beautiful"

Step 1: Term Frequency (TF) Calculation

We compute **TF** for the words **"sun"** and **"bright"**

Word	Doc 1 (5 words)	Doc 2 (6 words)	Doc 3 (7 words)	Doc 4 (6 words)
"sun"	1/5 = 0.2	0/6 = 0	1/7 = 0.1429	0/6 = 0
"bright"	1/5 = 0.2	1/6 = 0.1667	0/7 = 0	1/6 = 0.1667

Step 2: Document Frequency (DF)

DF is the number of documents that contain the word.

Word	DF (Number of Docs Containing Word)
"sun"	2 (Doc 1 & Doc 3)
"bright"	3 (Doc 1, Doc 2, Doc 4)

Step 3: Inverse Document Frequency (IDF)

Using the formula:

Using the formula:

$$IDF(t) = \log\left(\frac{N}{df(t)}\right)$$

Where **N = 4** (total documents)

$$IDF(\text{"sun"}) = \log\left(\frac{4}{2}\right) = \log(2) = 0.693$$

$$IDF(\text{"bright"}) = \log\left(\frac{4}{3}\right) = \log(1.33) = 0.176$$

Step 4: Compute TF-IDF

$$TF-IDF = TF \times IDF$$

Word	Doc 1 (TF × IDF)	Doc 2 (TF × IDF)	Doc 3 (TF × IDF)	Doc 4 (TF × IDF)
"sun"	$0.2 \times 0.693 = 0.1386$	$0 \times 0.693 = 0$	$0.1429 \times 0.693 = 0.099$	$0 \times 0.693 = 0$
"bright"	$0.2 \times 0.176 = 0.0352$	$0.1667 \times 0.176 = 0.0294$	$0 \times 0.176 = 0$	$0.1667 \times 0.176 = 0.0294$

What We Can Infer from TF-IDF Scores

1. **"Sun" has a higher IDF score (0.693) because it appears in fewer documents.**
   - It is more **unique** to specific documents, making it important for classification.
2. **"Bright" has a lower IDF score (0.176) because it appears in 3 out of 4 documents.**
   - It is **less useful** in distinguishing documents.
3. **Higher TF-IDF means a term is more relevant in that document.**
   - **"Sun" in Doc 1 & Doc 3 has a significant weight.**
   - **"Bright" is common and has low distinguishing power.**

Conclusion

- **Rare words get higher IDF scores, making them important for classification.**
- **Common words (like "bright") have lower IDF scores, meaning they are less useful.**
- **Search engines and NLP models use TF-IDF to rank words and documents.**